HipMRI Segmentation: Feasibility Review

COMP3710 | s4910001 | 23 October 2026

Project aim. My project tests whether a 3D Improved U-Net improves prostate segmentation at difficult end regions compared with a 2D U-Net, and whether better results are worth the extra computing cost. The output is a draft organ outline for a person to check.

Method and course concepts. Difficulty level: Hard. The main model is a 3D Improved U-Net: 3D residual units, encoder-decoder skip connections and deep supervision that adds predictions from different scales. Course concepts include convolutions, normalization and gradient-based learning. For both models, data preparation halves the depth, height and width (downsampling factor: 2). Encoder feature channels by level: 16, 32, 64, 128; batch size: 1 MRI volume; Adam learning rate: 0.001; random seed: 3710; loss: equal-weight cross entropy plus foreground Dice. The 3D dropout probability is 0.3. Model reference: Isensee et al., arXiv:1802.10508 (task reference [2]).

Success criteria:

1. Keep patients separate across training: 28 patients, 173 scans; validation: 5 patients, 19 scans; test: 5 patients, 19 scans. Save checked inputs, settings, file hashes and model checkpoints.
2. Target original test prostate scan-mean Dice >= 0.70 using the same prepared image grid; report all foreground Dice/IoU and patient means. Select the checkpoint with the lowest validation loss; choose the earliest if tied. This is how I apply the task's Dice target.
3. Use the same data splits and image grids. Compare end regions (source-K terminal fifths) and errors between nearby classes. These regions approximate the ends; they are not confirmed anatomical base/apex labels.
4. Use <= 4 GiB peak allocated tensor memory and <= 2 seconds median forward-pass time across scans after warm-up per prepared volume on one A100; measure preparation separately.
5. Compare 3-5 failure cases side by side, including end-region errors and separate predicted pieces. Recommend a model based on quality and cost. Five cases are shown; separate pieces in 3D still need checking.

Current results. Original fixed test (epoch 8): 5 patients, 19 scans; prostate scan-mean Dice 2D: 0.8411; 3D: 0.8354, meeting the project target. Patient-mean Dice is 2D: 0.8067; 3D: 0.8242; terminal-union patient-mean Dice is 2D: 0.6697; 3D: 0.7531. Scan means give more weight to patients with more scans. The 3D advantage at the ends does not appear in every split: original validation terminal patient-mean Dice is 2D: 0.7091; 3D: 0.6955 (checked on CPU).

Additional three-fold cross-validation leaves out the five test patients: 33 development patients, 192 scans, with 22 patients training and 11 evaluated per fold. Prostate patient-mean Dice improves from epoch 8 (2D: 0.6996; 3D: 0.7071) to epoch 12 (2D: 0.7657; 3D: 0.7546). At epoch 12, 2D has higher overall scores for all five foreground classes. Cross-validation ran after the original test. We chose longer training after seeing the epoch-8 cross-validation results. These extra experiments do not replace or expand the original test.

Earlier speed and memory test (epoch 4, same A100): median forward time 2D: 13.36 ms; 3D: 49.96 ms; allocated memory peak 2D: 445.89 MiB; 3D: 565.98 MiB. Prepared volume dimensions (depth, height, width): 64 x 128 x 128 voxels. Both meet the speed and memory targets. Times exclude data preparation, transfer and output processing. These measurements are not for the epoch-12 cross-validation models or memory used during training. Five pictured epoch-8 CPU validation cases show missed end regions, missed parts of the bladder and extra areas labelled as prostate; they are not from epoch-12 cross-validation or the test set.

Limits, next steps and backup plan. Limits include five test patients, one seed, similar scans from the same patient, shared training patients across folds, model design differences, downsampling and small CPU/CUDA differences. Next, I plan to rerun the fixed epoch-8 validation predictions and count separate 3D pieces and wrongly predicted islands: one CPU job limited to 15 minutes, 0 additional GPU minutes, no retraining. I also plan to run more training and evaluation tasks for both the 2D and 3D models to compare segmentation quality and result stability. These tasks have not started; computing time limits will be set before submitting the jobs. I would keep 2D as the faster backup and 3D as the main model being studied. Better results in some regions do not mean 3D is always better.